Field Deployment Guide: GX-7 Fibre Gateway

Applies to firmware 4.2.x · Revision C · Approved 9 October 2026

1 Purpose and Scope

This guide describes how field technicians install, provision and verify the GX-7 fibre gateway at residential and small business premises. It covers the physical installation, the provisioning file, the first boot and the acceptance tests that must pass before a job is closed.

Before you start

- Confirm the work order shows an active GPON port.
- Check that the optical light level at the wall box is between -8 dBm and -27 dBm.
- Make sure the customer is present to choose the gateway location.

Tools required

- Optical power meter with SC/APC adapter
- Laptop with a terminal client and an Ethernet port
- Fibre cleaning pen and inspection scope

2 Physical Installation

Mount the gateway on a wall or shelf at least one metre away from microwave ovens and cordless phone bases. Keep the fibre patch lead bend radius above 15 mm at all times.

1. Clean and inspect the SC/APC connector on the patch lead.
2. Connect the patch lead to the PON port on the rear panel.
3. Connect the power adapter and wait until the PWR light is steady green.
4. Check that the PON light turns steady green within three minutes.
5. Record the gateway serial number and MAC address on the work order.

Port	Label	Default use	Notes
PON	Green SC/APC	Fibre uplink	Class B+ optics
LAN 1	2.5G	Customer router or PC	Supports 2.5GBASE-T
LAN 2–4	1G	Customer devices	Bridged to LAN 1
TEL 1–2	RJ11	Voice lines	Disabled unless voice is ordered
USB	USB-C	Field diagnostics	Read-only storage

3 Provisioning

The gateway downloads its provisioning file from the auto-configuration server during first boot. If the download fails, the technician can load the same file locally from the laptop.

3.1 Provisioning file

A typical provisioning file for a residential internet and voice service looks like this:

```
# gx7-provision.yaml
device:
  model: GX-7
  firmware: 4.2.3
wan:
  mode: gpon
  vlan: 101
  pppoe: false
lan:
  subnet: 192.168.20.0/24
  dhcp_range: [192.168.20.100, 192.168.20.199]
voice:
  enabled: true
  lines: 1
  sip_server: sip.core.brightline.example
```

3.2 Service VLANs

VLAN ID	Service	Priority		Tagged on
		802.1p	Queue	
101	Internet	0	Best effort	PON
102	Voice (SIP)	5	Expedited	PON
103	IPTV multicast	4	Assured	PON, LAN 4
199	Management	6	Network control	PON

3.3 Loading the file locally

Connect the laptop to LAN 1, open a terminal session to the gateway and enter the commands below. The gateway reboots automatically when the commit succeeds.

```
gx7> login field
Password: ********
gx7# config load usb:/gx7-provision.yaml
gx7# config validate
Validation passed: 0 errors, 1 warning
gx7# config commit
Committing configuration... done
gx7# reboot
```

> Never commit a provisioning file that reports validation errors, because the gateway may lose its management VLAN and require a site revisit.

4 Acceptance Tests

All acceptance tests must pass before the job can be closed in the workforce application. Results are uploaded automatically when the technician presses Finish.

Test	Command or method	Pass criterion
Optical level	show pon optics	Rx between -8 and -27 dBm
Internet throughput	speedtest --server nearest	At least 90% of the subscribed rate
Voice call	Test call to 1505	Clear audio both ways for 30 seconds
Wi-Fi coverage	Walk test with the app	Signal above -67 dBm in every main room

The optical level command prints output similar to the following:

```
gx7# show pon optics
  Rx power      : -18.4 dBm
  Tx power      :   2.1 dBm
  Temperature   :  41.0 C
  Status        : IN RANGE
```

5 Troubleshooting

Symptom	Likely cause	Action
PON light flashing red	No optical signal or dirty connector	Clean and inspect the connector, then measure the light level
PON green, no internet	Wrong WAN VLAN	Check that wan.vlan is 101 and reload the file
Voice line silent	SIP registration failed	Run show voice status and call the network desk
Slow Wi-Fi	Channel congestion	Run wifi scan and move to the least used channel

Escalate to the network operations desk if a fault cannot be cleared within 30 minutes on site.

6 Revision History

Revision	Date	Change
A	12 March 2026	First issue
B	30 June 2026	Added IPTV VLAN 103
C	9 October 2026	Firmware 4.2.x commands and new acceptance tests